

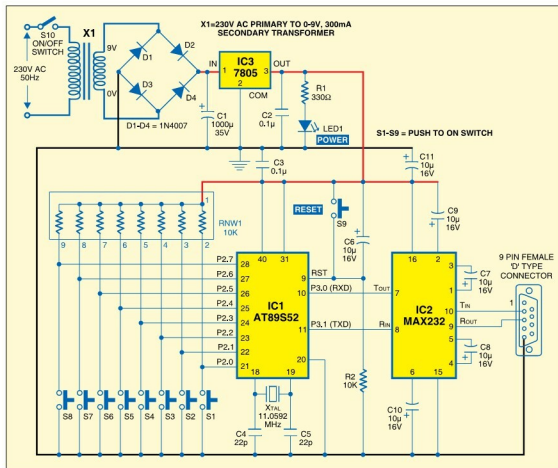


Fig. 2: Circuit for microcontroller-based serial data transfer to a PC

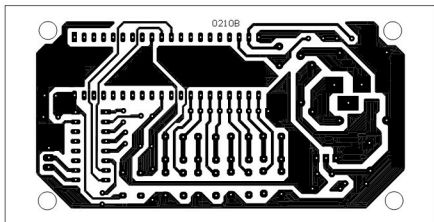


Fig. 3: Actual-size, single-side PCB for microcontroller-based serial data transfer circuit

Circuit description

Fig. 2 shows the circuit for microcontroller-based serial data transfer to a PC. It comprises microcontroller AT89S52, RS-232 driver MAX232, regulator 7805 and a few discrete components. The microcontroller detects

the switch pressed and the message is transmitted to the computer through the RS-232 driver.

The AT89S52 is a low-power, high-performance CMOS 8-bit microcontroller. It has 8 kB of Flash, 256 bytes of RAM, 32 I/O lines, a watchdog timer,

two data pointers, three 16-bit timers/counters, a six-vector two-level interrupt architecture, a full-duplex serial port, and on-chip oscillator and clock circuitry. In addition, the AT89S52 is designed with static logic for operation down to zero frequency and supports two software-selectable power-saving modes. The idle mode stops the CPU while allowing the RAM, timers/counters, serial port and interrupt system to continue functioning. The power-down mode saves the RAM contents

but freezes the oscillator, disabling all other chip functions until the next interrupt or hardware reset.

Port pins P2.0 through P2.7 are pulled high by resistor network RNW1 and also connected to switches S1 through S8, respectively. An 11.0592MHz crystal is used for providing the basic clock frequency. All I/O pins are reset to '1' as soon as RST pin goes high. Holding RST pin high for two machine cycles while the oscillator is running, resets the device. Power-on reset is derived from resistor R2 and capacitor C6. Switch S9 is used for manual reset.

The microcontroller provides a transmit channel and a receive channel for serial communication. The transmit data pin (TXD) is specified at port pin P3.1, and the receive data pin (RXD) is at port pin P3.0. RXD and TXD pins of the microcontroller are connected